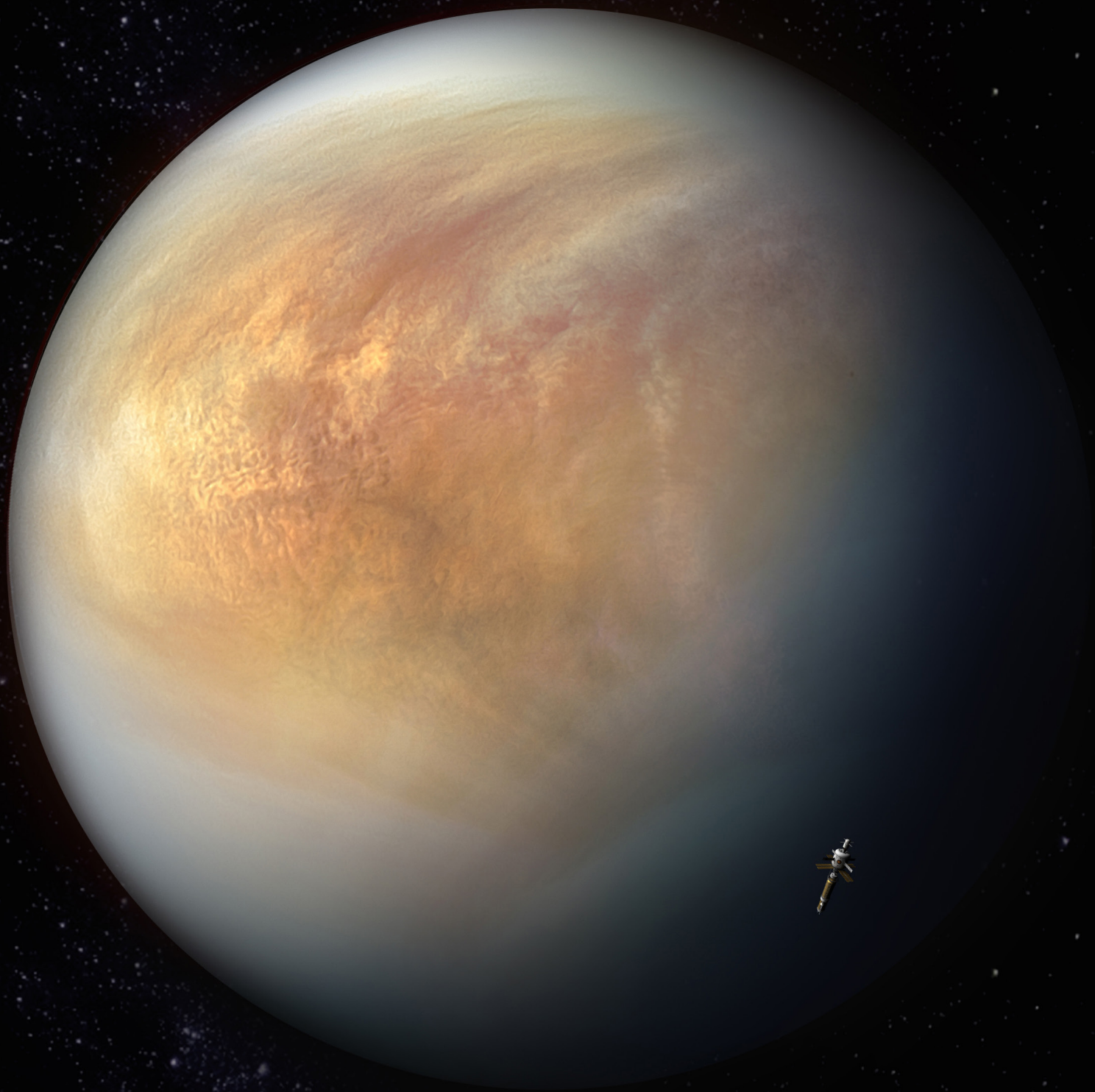


HIGH FRONTIER

4 ALL | BY PHIL EKLUND



SPACE DIAMONDS

INTRODUCTORY GAME BY JON MANKER

A game of space exploration for 2 to 5 players
Updated Dec 11, 2020.



iA. Introduction

Space Diamonds is designed to be your first game of *High Frontier*, introducing you to how Sails and Rockets move and getting you comfortable with the game's "subway map" representation of our solar system. The *Space Diamonds* rules are formatted in the same way as all other rules in *High Frontier*. You will play using only a portion of the Core box components (listed in **iB1**). There are two ways to play: an easier game where everyone uses Sails, or a bit more challenging game in which everyone uses Rockets.

- Sails.** Everyone travels using the same type of photon Sail that navigates space slowly but without water (fuel). Ignore the Rocket rules, which are written in blue font.
- Rockets.** Everyone travels using a Nuclear, Laser or Electric Rocket. You will travel faster than using Sails, but will need to concern yourself with Fuel management.
- Core Differences.** Rules that differs between *Space Diamonds* and *Core* are marked with a ♣ icon.

Core Game Variant. The discovery chits and mission cards of *Space Diamonds* can be added to any version of the *High Frontier* game (Core, modules, the **V11** variant, etc.). These provide another way to end the game.

iA1. ♣ Winning

The player with the most victory points (VP) at the end of the game is the winner. VP is awarded for winning the race back to LEO (Low Earth Orbit, the start point), and by accumulated discovery chits and completed mission cards.

- End of Game.** Once a player returns to LEO, he may not move any more. The game ends when all players have returned to LEO, or 3 Turns after any player has returned to LEO, whichever comes first.

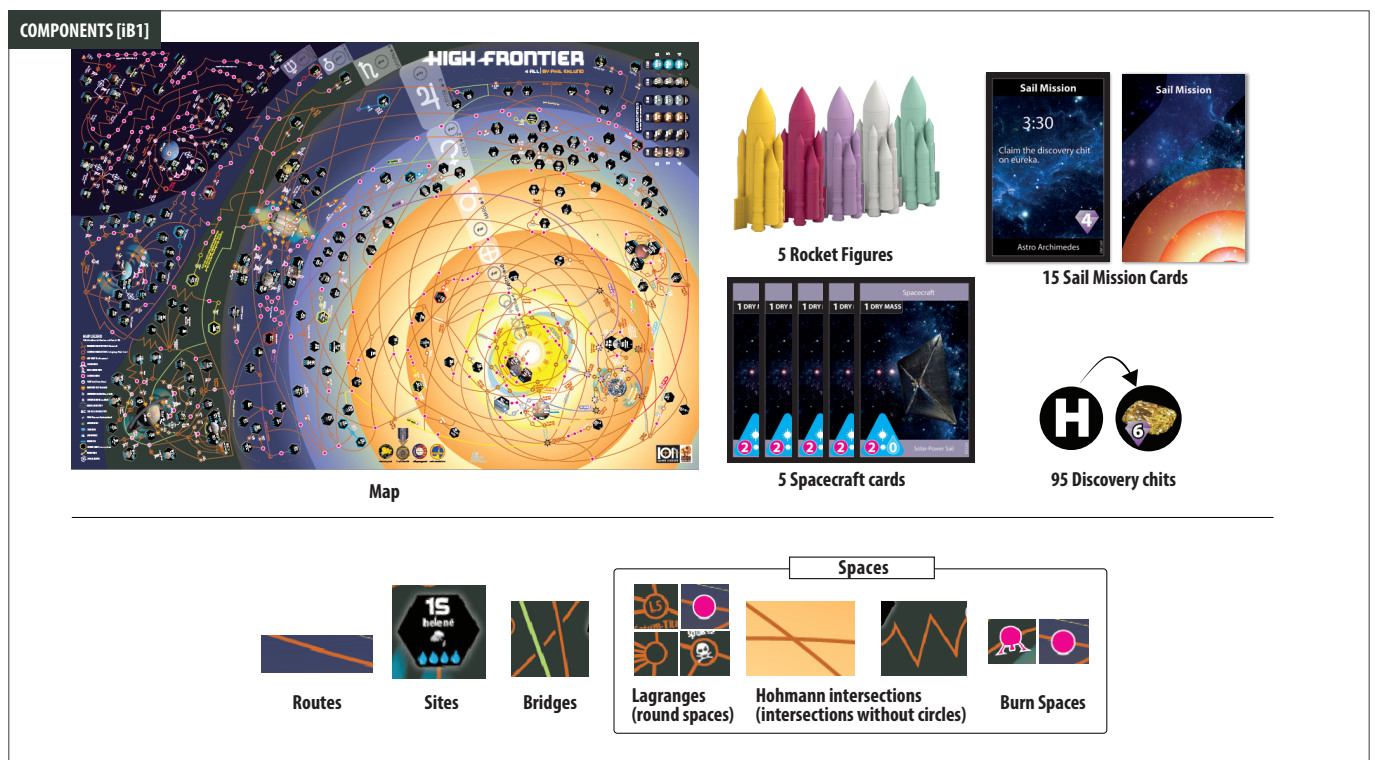
iB. Components & Anatomies

The *High Frontier* map, cards and playmats are used in *Space Diamonds*, but in a simplified manner. Additionally, *Space Diamonds* uses unique components called discovery chits (**iB2**). Below is a list of all components used in *Space Diamonds* (**iB1**).

- Heliocentric Zones** divide the map into concentric areas, each named after a planet: Mercury (☿), Venus (♀), Earth (♁), Mars (♂), Ceres (♁), Jupiter (♃), Saturn (♄), Uranus (♅), and Neptune (♆). The Sol clockface found around Sol on the map, in combination with the Heliocentric Zone, provides a way to locate Sites and objects on the map.
- Sites** are black hexagons indicating potential landing spaces.

iB1. Component List

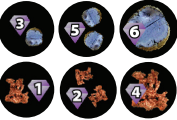
- The map
- 5 Rocket Figures, 1 in each of the 5 player colors
- ♣ 95 discovery chits
- ♣ 30 mission cards (15 Sail Mission, 15 Rocket Mission)
- ♣ 5 Spacecraft cards, each with a Sail side and a Rocket side.



iB2. Discovery Chits

Discovery chits are collected by the discovery operation and remain facedown both while on the map and in a player's possession. They are revealed to all only when they are taken from the map by a player.

- a. Chit Site Location.** The back side of the chits describes a Spectral Type (**C**, **D**, etc.) that is solely used to determine their starting location during **setup (iC2)**.
- b. Chit Type.** Chits are further divided into the following types: gem, ice, isofuel, science, life, and industrial.
- c. Effects.** Chits are worth victory points (VP) if you have them during endgame scoring, (the number in the chit). Some Chits can also be spent to gain **Bonus Burns (iE3b)** or to **add water to your Fuel Strip (iE6f)**.

DISCOVERY CHITS [iB2]	
 Backside (determines starting location during setup).	 Science
 Gems	 Life
 Isofuel (if discarding chit, gain 2 Bonus Burns).	 Ice
 Industrial	 Victory Points  If discarding chit, +1 tank of water to Fuel Strip.

iB3. Additional Components If Using Rockets

- a.** 5 Playmats
- b.** 5 Rocket Dry Mass Chits
- c.** 5 Rocket Wet Mass Chits
- d.** 5 additional Rocket Figures in 5 player colors (to indicate net thrust)

ADDITIONAL COMPONENTS [iB3]	
 5 Playmats	 15 Rocket Mission Cards
 5 Additional Rocket Figures	 5 Dry Mass Chits  5 Wet Mass Chits

iC. Setup

iC1. Sails or Rockets?

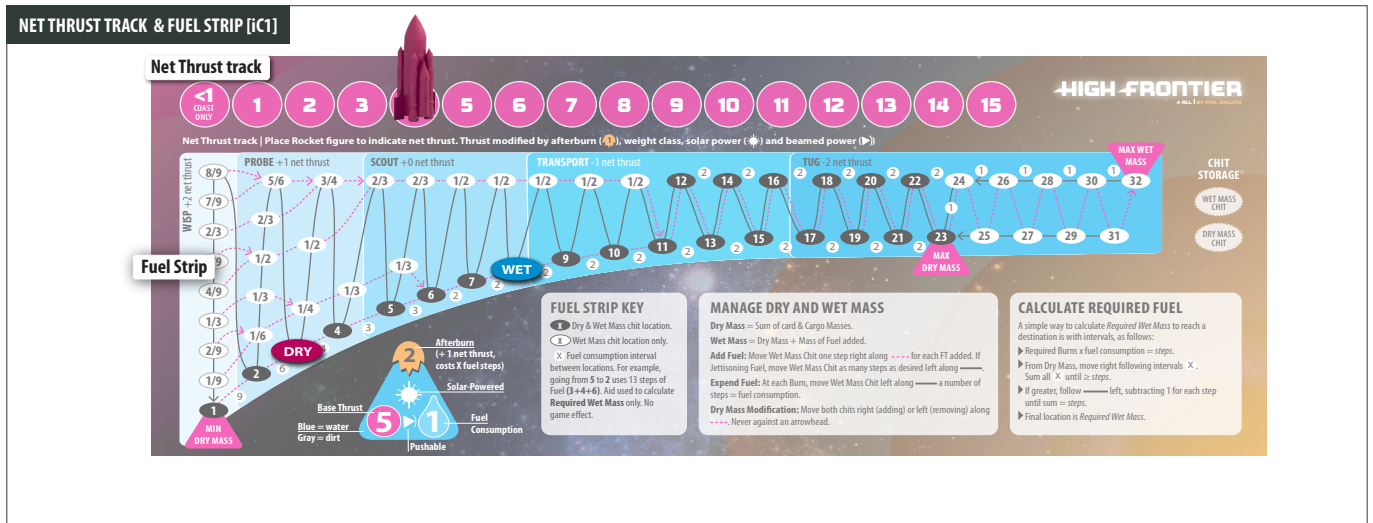
First, decide if your group will be playing with Sails or Rockets.¹ If using Sails, ignore all setup and rules in **blue** font. Then:

- a. Starting Location.** Each player takes a Rocket Figure of their chosen color and places it on LEO (the Space closest to Earth).
- b. ♣ Starting Mission Card.** Randomly deal each player a Sail mission card (**iG**). **If using Rockets, additionally deal each player a Rocket mission card.** Don't reveal your mission cards to others.
- c. ♣ Starting Spacecraft Card.** Randomly assign each player a Spacecraft card. Place your card faceup in front of you, showing the Sail side (the 2•0 thrust triangle) if using Sails **or the Rocket side (a 3•1, 4•2 or 5•3 thrust triangle) if using Rockets.**
- d. Fuel Strip.** Each player takes a playmat.

❗ EASILY MISSED: If playing with Rockets you start with one Sail and one Rocket mission. In this case, the Sail mission is accomplished with your Rocket. [iC1b]

¹ SAILS & ROCKETS. A sail is a vehicle that maneuvers using external forces, rather than expending any part of itself as does a rocket. This sacrificed portion, called **propellant**, provides the reaction mass for an action-reaction drive. Note that a rocket needs both **energy** (provided by fuel) and **mass** (provided by propellant) in order to maneuver. In space, which is full of energy but empty of mass, it is mass that is the most valuable. The propellant used in the game is assumed to be water.

- e. Wet and Dry Mass.** Note the Dry Mass number listed on your Spacecraft card. Place your Dry Mass Chit on the Fuel Strip section of your playmat indicating that number. Then all players place their Wet Mass Chit on the number 8.
- f. Net Thrust Indicator.** The first number in your Rocket's thrust triangle is your base thrust (**iE1b**). At the start of the game, place your second Rocket Figure on your base thrust number on the net thrust track at the top of the Fuel Strip.



iC2. Discovery Chit Placement

- a. ♣ Discovery Chits.** Mix these with their Spectral Type letters faceup; this is their back side. The front side of all chits should remain hidden during setup.
- b. ♠ Cover Sites With Chits.** The hexes on the map are called Sites. Each Site has a letter associated with it, called its Spectral Type.² Place a single discovery chit on each Site in the Mercury ♀, Venus ♀, Earth ⊕ and Mars ♂ Heliocentric Zones. Make sure that the Spectral Type letter of the chit matches the letter on the Site. If you are using Rockets, also cover the Sites in the Ceres Heliocentric Zone.
- c. ♣ Leftovers.** Place all remaining discovery chits back in the box, out of play, without looking at their fronts.

iC3. Player Order

Randomly determine the 1st player. Players take Turns clockwise.

iD. Sequence of Play

Start the game with the 1st player taking a Turn. Proceed taking Turns clockwise until one player triggers the end-game by returning their Spacecraft to LEO (**iH1**). Each of the remaining players take 3 more Turns, or earlier if they stop their Turns by reaching LEO. After these 3 Turns the game ends and the player with the most victory points (**iH2**) is the winner.

iD1. A Player's Turn

On your Turn, do either or both of the following, in any order:

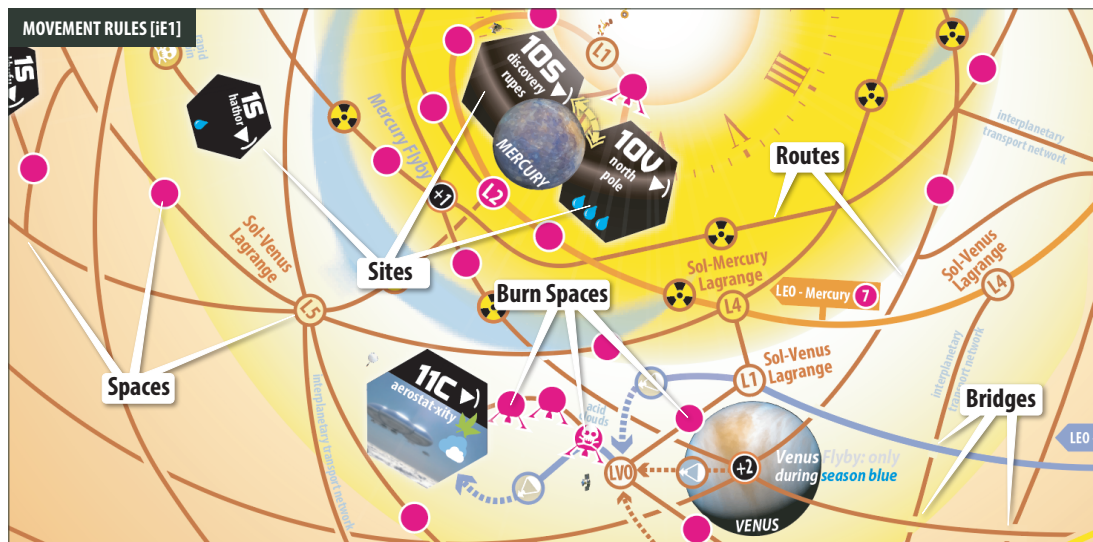
- Move** your Rocket (**iE**).
- Perform** 1 Operation (**iF**).

iE. Movement and Net Thrust

To move a Sail or Rocket:

- Calculate your **net thrust** which is your **base thrust** (**iE1b**) +/- modifications (**iE2**).
- This net thrust is how many **Burn Spaces** (**iE1g**) you can move on the map this Turn. Unused Burns cannot be saved from turn to turn.
- If you encounter or generate **Bonus Burns** (**iE3**), these are not deducted from your number of Burns remaining.
- If you follow a route without turning, or if you do turn but it is at an intersection with a round icon, then nothing is deducted from your number of Burns remaining.
- You must roll a risk if you enter a Space with a skull (**iE4**).
- Each Burn Space entered lowers your Wet Mass Chit by a number of steps on the Fuel Strip equal to your **fuel consumption** (**iE1m**).

² SPECTRAL TYPE. Asteroids are classified by the spectral analysis of their colors.



iE1. Movement Rules on the Map and the Cards

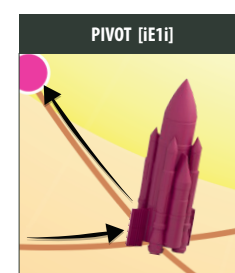
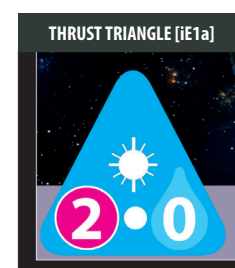
- Thrust Triangle** is found on each Sail or Rocket card.
- Base Thrust** is the left number in the thrust triangle. The larger the number, the more Burn Spaces you have the potential to move through with your Rocket Figure each turn.
- Net Thrust.** Your base thrust modified by different factors to/from your net thrust (i.e. base thrust as modified by **iE2**). The net thrust tells you what the maximum number of burns you can move through in a specific turn. However, you are not forced to use all the Burns you are granted by your net thrust or Bonus Burns.
- Routes** are the solid lines on the map you follow when you move your Rocket Figure. (You never move along the dotted lines in *Space Diamonds*).
- Spaces** are route intersections and sharp corners on routes, including intersections covered by a round icon. You may end your movement at any time in any Space. When exiting a Space, you can continue in any direction except the direction you entered from that Turn. The intersections without icons and the sharp corners are also known as **Hohmanns**. The intersections marked with a round icon are called **Lagranges** (ignore any text inside the circle).³
- Bridge** is any intersection where the routes don't touch. It is not a Hohmann and does not count as a Space in which you can change direction or end movement.
- Burn Space** is any Space filled with magenta color. They have various shapes but in *Space Diamonds* the shape does not matter. It costs 1 Burn to enter a Burn Space. When you have zero Burns left during your Turn, you cannot enter a Burn Space.⁴
- Changing Direction.** When exiting any Lagrange (**iE1d**), you may change your Rocket direction without spending any Burns (though you may have spent a Burn to enter the Space (**iE1f**)). When exiting a Hohmann, it costs no Burns if you follow the route without turning. If you do turn, see next bullet.
- Pivot** is a change of direction at a **Hohmann** (**iE1d**) within a single Turn. Both changing direction at an intersection and continue your movement through a sharp corner (zigzag) counts. It costs 2 Burns to make a Pivot.
- Directional Freedom.** At the start of your movement you may leave a Space in any direction without using Burns, regardless of what kind of Space it is, including the direction you came from in the previous Turn. (If you enter a Burn Space in your first movement step it will cost 1 Burn normally).

EXAMPLE [iE1g]

Your Sail starts in LEO on Turn 1. Your first step is always to move to the Burn Space labeled "cyclor". You decide to move to GEO, which expends your second and final Burn. Now you are coasting. Following route green, you pass through L1 and L5, halting in the next Hohmann because you have no more Burns which you need to enter the Burn Space above phobos. Note this trip took you through 5 Lagrange Spaces, in which you can freely change direction. For your second Turn you can set your Sail in any direction from your starting point. You choose to turn slightly left, passing 6 Lagranges: L4s, Burn Space, L3, Burn Space, L3, mjölnr orbit. You end your move by landing on the small stony asteroid mjölnr.

EXAMPLE [iE1j]

Your Sail starts in LEO. Following route red, it travels through cyclor, GEO, and L2 and stops at the red right angle turn (Hohmann). Since it used up both its Burns, it cannot travel any farther along route red. On its next Turn, it continues along route red towards Mars. Note that because of directional freedom, this does not count as a Pivot.



³ LAGRANGE POINT is a location in space where the combined gravitational forces of two larger bodies, for example Earth and Sol, equal the centrifugal force felt by a much smaller third body (e.g. a spacecraft). The interaction of the forces creates a point of equilibrium where a spacecraft can stay or turn almost effortlessly. —Joseph-Louis Lagrange, *Three-Body Problem*, 1772.

⁴ BURN SPACES. Why do only Burn Spaces count? Why doesn't every space cost movement points, like every other game? Because the game map was designed using delta-v maps of the solar system, it is a map of energy, not distance. "Delta-V" is shorthand for saying the change of velocity required to go to places in space. For instance, Earth orbits at a speed of 30 km/sec (km/sec = "kilometers per second"), but Mars orbits slower, only 24 km/sec. So the delta-v between Earth and Mars is at least 6 km/sec. Since each Burn in the game is a delta-v of 2.5 km/sec, one must traverse at least 2 Burns to get to Mars, plus extra to escape Earth.

REMINDER: The only time you spend Burns is when you enter a magenta space or when you are pivoting. (iE1I)

- k. Coasting.** A Spacecraft is allowed to keep moving, even if it has no Burns left, has a net thrust of zero, or has ran out of Fuel, provided it does not enter any Burn Spaces or make any Pivots. A coasting Spacecraft can even land on Sites, spend accumulated Bonus Burns or pick up additional **Bonus Burns (iE3a)**. Remaining Burns are lost and not saved for the next Turn.
- l. ♣ Landing Sites.** Hexagonal Spaces are Sites where you can land, representing the moons, comets and asteroids in our solar system. Some large planets and moons have more than 1 Site. To land on or liftoff from a Site does not cost any extra Burns but if you land it ends your movement for that Turn. Each Site has a number indicating its Size, but this is not relevant while playing *Space Diamonds*.

EXAMPLE (iE1I) Your Sail starts in GEO near Earth, bound for the martian system. Following route red, it travels through L2 and Pivots at the red right angle turn (Hohmann). This costs both of its Burns. It coasts along route red, to the L2 Sol-Mars Lagrange, finishing its move with a landing on deimos, a small moon of Mars.

m. Fuel Consumption is the right number in your thrust triangle. It indicates how many steps of water you consume for each Burn (iE6). The lower the better. Sails move due to pressure from sunlight and solar wind, so their fuel consumption is zero.⁵

iE2. Modifications to Base Thrust = Net Thrust

The base thrust is what is printed on the Sail or Rocket card. For Sails, this is modified by the Solar-Power modifier (below). For Rockets, this is modified by the weight class modifier (below).

- a. Solar-Power Modifier (Sails Only).** Each Heliocentric Zone (iBa) depicts on the map a modifier used to determine the net thrust of Sails.⁶ Use the modifier of the zone in which you start your movement: +2 for Mercury ☿, +1 for Venus ♀, +0 for Earth ⊕, -1 for Mars ♂, -2 for Ceres ♁ and even less for zones further out. This makes it very hard to move outside the Mars Zone with a Sail. There are no discovery chits beyond Ceres, so you will find little reason to travel there beyond Refueling, some Missions or the occasional shortcut.

EXAMPLE (iE2a) Your Sail has a base thrust of 2, and starts its move in the Mars Zone. Its net thrust is therefore one.

- b. Weight Class Modifiers (Rockets Only).** Increase or decrease the net thrust by the amount indicated by the location of your Wet Mass Chit: wisp (+2), probe (+1), scout (+0), transport (-1), tug (-2).

EXAMPLE (iE2b) Your 4-2 Rocket starts as a scout, with a weight class modification to its net thrust of +0. You burn 9 steps of water which reduces your weight to probe, which obtains +1 net thrust. On your next Turn your net thrust is 5 instead of 4. Afterward, you refuel (iF3) 4 times on a comet, bringing your Wet Mass Chit to 20. This is a tug, reducing the net thrust to 2.

iE3. Bonus Burns

As you move, you may move beyond the **burn limit (iE1c)** of your net thrust in two ways: entering flybys or discarding isofuel chits. These generate optional Bonus Burns that cannot be saved to the next Turn. **Bonus Burns cost no Fuel!**

- a. Flybys.** Some planets, such as Mars, have a Space with a +X number and the text "Flyby". For instance, at Mars it says "Mars Flyby" and "+1" as the X. The X is the number of **Bonus Burns** gained by entering the Space. You may enter several Flybys during the same turn. Special case: The solar oberth flyby only grants Bonus Burns once per Turn.

EXAMPLE (iE3a) Your Rocket starts in GEO, and follows route blue through L2, a Burn Space (where it must spend Fuel), and the Mars +1 flyby. The Mars gravitational slingshot gives your Rocket a Bonus Burn which it uses to continue its voyage to Ceres.

- b. ♣ Isofuel** is a type of discovery chit. It has a yellow drop icon on it. You may keep this chit for VP, or discard it at any time to gain 2 Bonus Burns. These Bonus Burns work like flybys above. You may discard as many of these chits as you like during a single movement.

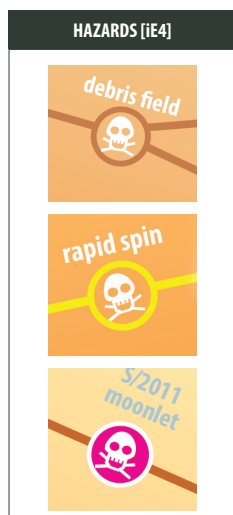
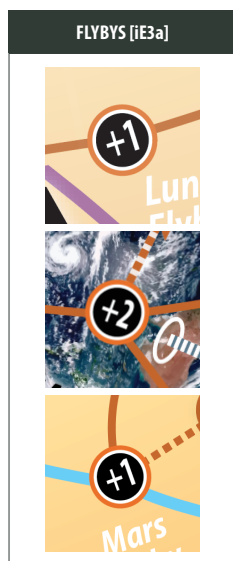
iE4. Hazards

These Spaces are indicated with a skull icon and can be found combined with or without a Burn Space. As soon as you enter a Space with a Hazard skull, pause your movement (after spending the Burn, if required) and make a Diamond Hazard Roll:

- a. ♣ Diamond Hazard Roll.** Roll a dice. If you roll a "1", immediately discard out of the game your choice of half your acquired discovery chits, rounded up. Additionally, you have to stop your movement for this Turn (to repair your ship). All remaining burns are lost, and you may not coast. If you roll a 2, 3, 4, 5 or 6 you may continue your turn as normal.

⁵ ROCKET FUEL CONSUMPTION is a measure of how fast the propellant is expelled out the nozzle. The faster, the more sparing the rocket is on its precious supply of water. But a rocket engineer faces an agonizing choice. A rocket has only so much combustion energy, which can be either given mainly to the propellant, expelling it faster for better water economy, or given to the vehicle itself, making it go faster with high thrust. Thus the tradeoff is between economy or thrust, just as a fuel-efficient car won't accelerate like an American muscle car. But in a car, even a fuel-efficient one, almost all the energy is spent moving the car. In a "fuel-efficient" rocket, almost all the energy is wasted accelerating propellant into the uncaring universe, forever unavailable to move the rocket itself.

⁶ HELIOCENTRIC ZONES show how far the Sail is from Sol. Because sails are pushed by solar energy, their net thrust diminishes as they get further out.



iE5. ♣ Emergency Sail

If at the beginning of your turn, your Sail has a net thrust is zero (or less), or if your Rocket has no Fuel, you are allowed to use an emergency Sail for your move. This movement, not used elsewhere in *High Frontier*, allows you to move one Space during a Turn. Remember, every intersection is a Space. You may not coast. You are allowed to land on Sites.

iE6 Rocket Fuel Strip

Rockets differ from Sails in that they track Fuel.

- Wet and Dry Mass Chits** are placed on the Fuel Strip on the playmat to track your Rocket's weight and Fuel. Mark the Rocket's Dry Mass (weight without Fuel)⁷ and the other marks Wet Mass (weight including Fuel). Dry Mass never changes in *Space Diamonds* (iC1e), but the Wet Mass changes as you add or subtract water (Fuel).
- Adding Water** moves the Wet Mass Chit a number of steps to the right, following the red dotted line. See iE6f and iF3.
- Spending Water** only occurs when spending a Burn and moves the Wet Mass Chit a number of steps to the left, following the black line. This means that you sometimes only spend part of a tank of water for a Burn. Adjusting the Wet Mass Chit may change your *weight class* (iE2b).
- Movement Procedure.** Every time you spend a Burn move your Wet Mass Chit to the left a number of steps equal to your fuel consumption (the right number in the thrust triangle) for each Burn.

EXAMPLE [iE6d] A 5-3 Rocket has no modifications and thus a net thrust of 5. It enters 1 Burn Space and makes 2 Pivots. This spends $1+2+2=5$ Burns in 1 Turn, which is the maximum allowed. Each Burn will move the Wet Mass Chit 3 steps to the left following the black line. 5 Burns x 3 steps = 15 steps of water spent. If the Wet Mass starts at 8, 15 steps takes it down to 2-5/6.

- Out Of Fuel!** If your Wet Mass Chit moves on top of the Dry Mass Chit, your Rocket has no more Fuel and cannot perform any more Burns until it regains water it needs for Fuel. It can still coast (iE1k) until it starts a turn without fuel, in which case the *emergency sail* movement (iE1k) must be used, and no further coasting is possible. The Wet Mass Chit can never move to the left of the Dry Mass Chit, as this would indicate an impossible 'less than zero' amount of Fuel.

EXAMPLE [iE6e] A 5-3 Rocket has 2 steps of Fuel remaining. Since each Burn costs 3 steps, it is effectively "out of fuel".

- Chits Spent for Water.** If you have a discovery chit with blue drop(s) on it, you can discard it to gain Fuel on the Fuel Strip. At any time during your Turn, you may discard the chit and move the Wet Mass Chit 1 step to the right for each drop icon on the discovery chit, following the red dotted line.

EXAMPLE [iE6] At the start of the game, your 3-1 electric rocket has a Wet Mass of 8 and a Dry Mass of 3. Starting in LE0, it follows route blue through 3 Burn Spaces, then coasts to the Mars flyby. This gives it a Bonus Burn, enough to move to the blue Hohmann intersection in the Ceres Zone. There it stops, with not enough Burns remaining to take the Pivot. This move took the Rocket through 3 Burn Spaces, so it must expend its fuel consumption three times. This lowers its Wet Mass Chit 3 steps, along the black line, from 8 to 6-1/2. On Turn 2, it continues in a new direction, still following route blue. Note this does not count as a Pivot, because at the start of a move you can move in any direction. The Rocket flies through a Pivot and a Burn Space, ending its move at the Gefion Family barycenter. Wet Mass is reduced another 3 steps, ending at 5-1/3. On its third turn, the Rocket travels through a Burn Space (half Burns are the same as Burns in *Space Diamonds*) to land on Ceres. This drops its Wet Mass to 5, with 7 more steps of fuel before it is empty.

TIP: Water and Rocket Fuel are the same! [iE6]

EASILY MISSED: Fuel is added to your Rocket in tanks (following dotted red line), but is burned in steps (following black line). [iE6d]

iF. Operations

iF1. ♠ Discovery Operation

When your Spacecraft is located on a Site with a chit you may perform a discovery operation. Reveal the chit to all players and then place it facedown in front of you. You may look at your discovered chits at any time but other players only get to see them when you discover them.

iF2. ♣ Scanner Operation

Choose up to 3 Sites (anywhere) with discovery chits still on them. Secretly reveal these Site's Hydration (the number of water drops on it) by temporarily sliding the discovery chit aside and then back. You may not look at the hidden face of the chit.

iF3. ISRU Refuel Operation

When your Spacecraft is located at a Site, it may perform an ISRU Refuel Operation.⁸ Move the Wet Mass Chit rightwards as many steps as the Site's Hydration, following the red dotted lines.

⁷ WATER AS FUEL? Despite game terminology in which you can "refuel" with water at a site, the game treats water not as fuel but as propellant. Propellant provides mass, not energy. Although a real rocket needs both fuel and propellant to accelerate, the game tracks only water propellant, not fuel. A rocket normally needs far more propellant than fuel. A nuclear steam rocket, for instance, may need tonnes of water propellant but only grams of uranium fuel.

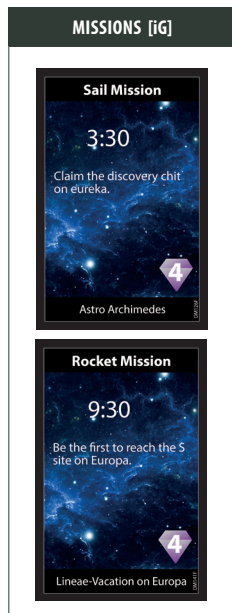
⁸ ISRU stands for In-Situ Resource Utilization. It means to "live off the land", using resources (such as water) discovered and extracted from space, rather than imported from Earth.

EXAMPLE (iF3) You start your Turn at a Site. You chose to make an ISRU Refuel Operation before you move. You end your movement on another Site with a chit on it. You must wait until your next Turn to take a discovery operation while still on that Site. It is possible for an opponent on their Turn to land and snatch the chit before you can.

iG. Missions

At the beginning of the game you received one or two mission cards, kept secret from the others. You can attempt one or both missions, each worth VP in endgame scoring.

- a. **Accomplishment.** At the moment you accomplish a mission, announce your success to the others and tuck the card under your playmat with the VP visible.
- b. **Locations.** Some missions direct you to perform a *discovery operation (iF1)* at a specific Site. These cannot be accomplished if someone else has already claimed that chit.



iH. Game End & Scoring

iH1. The Endgame Phase

You are not allowed to return to LEO until you have accumulated at least 4 *discovery chits (iB2)*. As soon as any player (with this many chits) returns to LEO, it triggers the **endgame phase**. This player will track 3 more full turns for the remaining players (you may find it helpful to use a marker on the net thrust track to count the turns).

- a. **Ticker Tape Parade.** During Endgame Phase, any player who reaches LEO takes no further turns.
- b. **Order.** Keep track of the order in which the players return to LEO, as this is relevant for scoring.
- c. **Game Ends** when all players have reached LEO or the 3 turns have passed. Proceed to endgame scoring.

iH2. Endgame Scoring (What Gives You VP?)

Each player sums up their victory points (VP), regardless of whether they completed their return trip to LEO. The VP consists of:

- a. **Race Home.** The 1st player to return to LEO = 4 VP, 2nd = 3 VP, 3rd = 2 VP, 4th = 1 VP and the 5th = 0 VP. If you did not return, receive 0 VP.
- b. **Discovery Chits VP.** Regardless if you return or not, receive the VP value of all your chits (**iB2**).
- c. **Completed Mission Card VP.** Regardless if you return or not, receive the VP value of your completed mission cards (**iGa**).

iH3. Winner

The winner is the player who has the most VP. In case of ties, the player with the most gem chit VP (**iB2b**) wins. Still tied? Share the victory (or position).



ii. Starship Builder Variant (courtesy Tom & Rick, 13 year old rocketeers).

This introductory cardgame for 2 to 6 players lasts a half hour, and gets you accustomed to the High Frontier cards. It requires understanding *supports (J)* and Operational (glossary).

ii1. Starship Builder Setup

Using only the 6 patent decks of the Core Game, setup per (**C4**). No other components are used.

ii2. Starship Builder Gameplay

- a. **First Round.** Randomly determine the first player. Clockwise everyone takes one card from one of the six patent decks.
- b. **Subsequent Rounds.** Start each round by moving the top patent cards to the bottom of their deck. The next player clockwise is the new first player, then continue clockwise taking one card for each player. There is no maximum Hand.
- c. **Endgame.** As soon as one of the decks runs out, each player assembles a rocket carrying an Operational thruster, robonaut, and refinery, and discards unwanted cards. Award each player a point if they have the best in one of 4 categories: lowest overall mass (**B2b**), highest thrust (**B2d**), lowest fuel consumption (**B2d**), and highest *rad-hardness*, counting the card with the lowest rad-harness, *see B2j*). Modify both thrust and fuel consumption by *movement-modifying supports (B2i,J5)* that are part of the *support chain (J1c)*.